MERS Filtering Equations (v30)

A companion to King, Lin & Ionides, arXiv:2405.17032

Contents

Disclosure. Prepared with assistance from Claude (Anthropic) and ChatGPT (OpenAI) for LaTeX typesetting, computation checks, and exposition. All mathematical content verified independently by the author against KLI (arXiv:2405.17032) and the `phylopomp` source.

Changes in v25. *(i) Notation (per A. King).* Both filters are now written as sums of their component terms: the singular filter is assembled as $w = \sum_{u\in\mathcal{U}_e(y)} S_u$ (§7.2), symmetric with the regular $\partial_t w = \sum_k I_k - (\text{outflow}) - \lambda w$ (§7.4); the six singular terms (Eqs. 1–6) are relabelled as the components $S_{\mathrm{SC}}, S_{\mathrm{SH}}, S_{\mathrm{TCC}}, S_{\mathrm{THH}}, S_{\mathrm{THC}}, S_{\mathrm{TCH}}$. *(ii) Demography.* Human death corrected to the constant rate B_H (camel/human symmetric), matching the shipped `mers.yml`; the v24 per-capita term $\frac{B_H}{N_H} S_H$ is removed.

Changes in v26 (review cleanup). (a) The assembled singular filter is now an *unnumbered* display, so the regular components keep their names Eqs. 7–12. (b) The compatible mark set is written $\mathcal{U}_e(y)$ (it depends on the post-event coloring), with its cases and the analytic fixed-destination update given explicitly. (c) The forward/SMC (parent-conditioned) enumeration is removed from §7.2 and left to §9/§11, so the analytic update cannot be misread. (d) The reduced regular exit rate (§8.2) spells out the demography term $B_C + B_H + B_C \mathbb{1}_{S_C>0} + B_H \mathbb{1}_{S_H>0}$ in place of "neutral". (e) The Eq. 7/8 "net" remark is softened (inflow uses x', outflow x). (f) The intro's duplicated metadata sentence is cut.

Changes in v27. (a) Added §2, a plain-language overview placed before any formal notation, so a reader can get the intuition for the filter and for the four proposal kernels (naive/soft/guided/hard) before the derivation. (b) The guided-family kernels subsection (§9.1, soft/guided/hard proposal laws and their support-safe ε-floor, added without a version bump in the prior revision) now uses KLI's own proposal symbol π throughout in place of an ad hoc q, and states explicitly which symbols in that subsection are this repository's vocabulary rather than KLI's. (c) Added a final §12, an implementation map tying every formula in §9.1 to the Julia code that implements it and to this repository's existing SEIR implementation, which the guided-family MERS kernels mirror. (d) Removed a second stray fragment that had been left after `\end{document}` (and so never compiled), duplicating and partially contradicting §9.1.

Changes in v28. *Demography, reverted to per-capita death (again).* `mers_naive.jl` was resynced against upstream `PhyloPOMP.jl` (`devel` branch), which rewrote its demography death rates from the constant, guarded form that v25 adopted back to the per-capita form v25 had removed: $B_C\, S_C/N_C$ and $B_H\, S_H/N_H$ (both demes, symmetric), in place of $B_C \mathbb{1}_{S_C>0}$ and $B_H \mathbb{1}_{S_H>0}$. The same change has been ported into `mers_soft.jl`, `mers_guided.jl`, and `mers_hard.jl` (previously still on the constant/guarded form) so all four kernels agree. *(Corrected in v30 — see below. The v28 text asserted here that this diverged from the R reference model; it does not.)* See §8.2 below for the updated reduced exit rate, and `mers_kernel_diagnostic_findings.md` for a related, separate correction (the real-tree collapse point under current defaults is the first Human sample, $\chi_H = 0$, not camel extinction — that finding is empirical/diagnostic and is not repeated in the derivation here).

Changes in v29. $\varepsilon = 0$ *is now a legal "no floor" mode.* `proposal_floor` (the ε of §10) previously required $0 < \varepsilon \le 1$; the lower bound is now $0 \le \varepsilon \le 1$. Every floor helper (`floored_shares`, `floored_shares_feasible`, `cross_proposal`, `floored_branch_law`, `hard_branch_law`) already reduced algebraically to the plain (un-floored) base law at $\varepsilon = 0$; only the validation changed, plus each helper's docstring now states this explicitly. This restores the option this document has argued against manufacturing: at $\varepsilon = 0$, a target-feasible outcome with a genuinely zero guide weight or relative hazard receives exactly zero proposal mass and the corresponding particle dies, rather than being kept alive by the floor's mixture term. The default remains 0.05 (support-safe); $\varepsilon = 0$ is opt-in, added specifically so a user who prefers to let this class of failure show up as $-\infty$ can do so without disabling the floor helpers' separate, unrelated all-zero-weight fallback (which still substitutes a uniform law when literally no weight information is available at all, independent of ε).

Changes in v30. *Correction: the v28 Julia/R demography "divergence" was an error and did not exist.* v28 asserted that the per-capita death rates $B_C S_C/N_C$, $B_H S_H/N_H$ disagreed with the R `phylopomp` reference `yaml/mers.yml`, and §8.1 repeated the claim. That conclusion was drawn from stale local clones of the R repository, which had not been fetched: their newest commit touching `yaml/mers.yml` was `817e558` (2026-07-02), predating the change in question. Upstream `kingaa/phylopomp` commit `f636a7e` ("fix MERS demography", 2026-08-01) rewrote `death_c/death_h` to `params.Bc/params.Nc * state.Sc` and `params.Bh/params.Nh * state.Sh`, and dropped the `if (S>0)` guards — exactly the form this repository's kernels use. **The Julia and R models agree on this term**; both were corrected in the same direction. Nothing in the derivation, the code, or the reduced exit rate changes as a result — only the erroneous divergence claim is withdrawn.

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1 Plain-Language Overview

This section is deliberately informal and does not aim for notational precision; the precise derivation begins in §2. Every technical term introduced here is either KLI's own (arXiv:2405.17032) or is explicitly marked as this repository's own vocabulary, not KLI's.

The data and the question. We are given a *genealogy*: a tree describing how a set of sampled MERS-CoV sequences are related, with each tip labeled by the host species it was sampled from (camel or human — this labeling is M, the tip metadata, in §2). We want the likelihood of this tree under a two-host epidemic model: how probable it is that an epidemic with given parameters ($\beta_{CC}, \beta_{HH}, \beta_{HC}, \beta_{CH}, \ldots$, defined in §3) would have produced exactly this genealogy.

The hidden part: coloring. The tree's branches do not, on their own, say which host species carried each *internal* (unsampled) lineage — only the tips are typed. KLI call the assignment of a deme (host species) to every branch the *coloring* Y; it is latent. Computing the likelihood exactly means accounting for every population history and every coloring consistent with the observed tree and its tip types — this is what the expectation $\mathbb{E}_\theta[\cdot]$ in KLI's likelihood formula (§2) means concretely.

Why we cannot just simulate forward. If we simulate the epidemic forward in time, choosing at each transmission event which host gets infected and, for cross-species transmission, which existing lineage (if any) changes color, most simulated colorings will not match what the real data requires — e.g. a lineage may be colored "camel" when the data needs it to end up "human". Rejecting every mismatched simulation outright would be extraordinarily wasteful. KLI's Theorem 5 instead licenses *importance sampling*: any consistent way of guessing the coloring is valid, as long as it never assigns exactly zero probability to an outcome the data allows, provided we track how far our guess differs from "the correct odds" and correct for the difference afterward with a multiplicative *importance weight* (KLI's boost B and decay λ, §9).

Proposal vs. target. KLI's term for "our guess" is the *proposal kernel* π (what we actually simulate from, §10); the "correct odds" implied by the true model is captured by the *compatibility factor* Φ (§3, §5). The weight that corrects one for the other is $B = \Phi/\pi$: exactly right when $\pi = \Phi$, but valid — just noisier — for *any* π that is positive wherever Φ is positive (KLI Theorem 5, Lemma 3). This single fact is what licenses everything in §10: we are free to choose π to make simulations land on the correct coloring more often, without changing the exact likelihood being estimated.

Four ways to guess: naive, soft, guided, hard. *These four names are this repository's own vocabulary; KLI's paper discusses the existence of a valid π in general but does not name specific choices of it.* This repository implements four:

- **Naive** guesses using only the current population counts: e.g. if 3 of 10 infected camels are lineages we are tracking, guess "no color change" with odds proportional to the 7 untracked camels, and "some tracked camel's color changes" with odds proportional to the 3 tracked ones. It never consults the data being fit beyond the current counts.
- **Soft, guided, and hard** all consult a cheap, once-per-genealogy forecast — the *filter guide* in PhyloPOMP, built from a simplified reverse-time approximation of the true dynamics — of which deme each lineage is heading toward, given what is eventually observed at the tips. They differ in *how much* say the guide gets:
  - **Soft** lets the guide pick *which* tracked lineage's color changes, but keeps the same odds as naive for *whether* a change happens at all.
  - **Guided** lets the guide influence *both* whether a change happens and which lineage it lands on, in one combined guess.
  - **Hard** lets the guide's opinion be strong enough to suggest color changes *faster than the epidemic itself would produce them*; the extra (or missing) rate is debited from (or credited to) a bookkeeping term called *decay* so the importance weight still comes out exactly right.

Why floor the guess (the "ε-floor"). *"ε-floor" and the symbols ε, π_ε, κ_ε used in §10 are this repository's own terms, not KLI's.* If the guide is very confident — say it assigns essentially zero chance that a particular lineage's color changes — but the true data actually requires exactly that change, then soft/guided/hard's guess would assign that outcome probability 0. Recall $B = \Phi/\pi$: if $\pi = 0$ while $\Phi > 0$, the weight is $\Phi/0$, undefined, and that particle is lost with no way to recover. To prevent this, every guided-family proposal mixes in a small guaranteed floor $\varepsilon \in (0, 1]$ of probability, spread uniformly over every outcome that is still physically possible, so no possible outcome ever gets exactly zero proposal probability. This does not change the exact likelihood being computed (by the $B = \Phi/\pi$ argument above) — it only changes how efficiently the simulation explores the possibilities.

Roadmap. §2–§9 build KLI's exact filter equation, independent of any proposal choice. §10 then derives the four proposal kernels precisely. Finally, §13 (after the summary) maps every formula in §10 onto the Julia code that implements it, including which pieces are direct analogues of this repository's existing SEIR implementation.

2 What King, Lin, and Ionides Establish

The problem. For the general colored problem one is given a pruned genealogy $P = (T, Z, Y)$ with n typed tips. KLI's obscuring operator discards all deme and inline-event information not visible from the tree topology: $\mathrm{obs}(P) = (T, Z)$, the *obscured* genealogy, with the internal coloring Y latent. Let M denote the camel/human tip metadata. Per KLI Remark 3, M is not part of (T, Z) itself; it enters through the sample-associated compatibility factors, which in the MERS specialization allow only the camel chop S_C at camel-labeled leaves and only the human chop S_H at human-labeled leaves. Given a parametric structured population model with parameters θ, the likelihood computed here is therefore the joint density of the obscured genealogy *and* the tip metadata,

$$L(\theta) = \Pr_{\theta}\{(T, Z), M\} = \mathbb{E}_{\theta}[\Pr\{(T, Z), M \mid H_T\}]\,,$$

where the expectation averages over all compatible population histories H_T. Prior approaches required either coalescent approximations, typically requiring large populations and negligible sample fractions, or linear birth–death models. KLI derive an exact likelihood construction for a broad class of discretely structured continuous-time Markov population models.

The population model. A CTMC on state space $\mathcal{X}$ with D demes $\mathcal{D} = \{1, \dots, D\}$. Each jump mark $u \in \mathcal{U}$ has a rate kernel $\alpha_u(t, x, x')$, a jump $x \to x'$, and a production vector $r^u = (r_1^u, \dots, r_D^u) \in \mathbb{Z}_{\geq 0}^D$, where r_d^u is the number of lineages produced in deme d at event u. Each jump mark may be of birth type (new lineages created), death type (lineages removed), migration type (lineages change deme), sample type (samples collected), neutral type (no genealogical change), or compound type, such as sample/death. The deme-occupancy function $n(x) = (n_1(x), \dots, n_D(x))$ counts active individuals in each deme.

Key quantities. For a pruned genealogy P with coloring Y and a population jump of mark u at time t:

- **Pruned lineage count** $\ell_d(Y_t)$: number of branches of P in deme d immediately *after* t (the post-event coloring).
- **Saturation** $s_d = s_d(Y_t)$: number of pruned lineages assigned to the r_d^u production slots in deme d at this event.
- **Binomial ratio:**
$$\phi_u = \prod_{d=1}^{D} \frac{\binom{n_d(X_t) - \ell_d(Y_t)}{r_d^u - s_d}}{\binom{n_d(X_t)}{r_d^u}}.$$
This is the conditional probability that exactly s_d of the r_d^u produced lineages are pruned, given the current occupancy. Both the occupancy $n_d(X_t)$ and the count $\ell_d(Y_t)$ are post-event.
- **Compatibility indicator** $Q_u(Y_{t^-}, Y_t)$: equals 1 iff the coloring transition $Y_{t^-} \to Y_t$ is consistent with mark u and saturation s (the operators $\sigma_{dd'}^{b}$ or $\kappa_{d\,d_b\,d_{b'}}^{bb'}$ match the slot structure), and 0 otherwise. This is the *only* place the pre-event coloring Y_{t^-} appears.

Theorem 1 (KLI Theorem 1, specialized to the present notation)**.**

$$\Pr[P_T = P \mid H_T = H] = \mathbb{1}_{\mathrm{ev}(H) \supseteq \mathrm{ev}(P)} \cdot \prod_{t \in \mathrm{ev}(H)} \phi_{U_t}(X_t,\, Y_{t^-},\, Y_t)\,,$$

where $\phi_u = 0$ whenever $Q_u(Y_{t^-}, Y_t) = 0$, and equals the binomial ratio otherwise. The product is over all population event times; events not involving a pruned lineage contribute their $s = 0$ ratio.

In $\phi_u(x, y, y')$, $x = X_t$ and $y' = Y_t$ are the post-event population state and coloring, respectively, with X and Y taken to be càdlàg. The lineage count and saturation entering the binomial ratio are therefore evaluated at Y_t, whereas the pre-event coloring $y = Y_{t^-}$ enters only through Q_u.

Appendix A of KLI proves the result by adding sampled lineages one at a time; each extends backward until it coalesces with an already-tracked lineage or reaches $t = 0$. At each population event the lineage either emerges from the event or does not; the product in Theorem 1 accumulates these local factors.

The filter of KLI Theorem 5. The data are the *obscured* genealogy $\text{obs}(P) = (T, Z)$: the tree-structure Z with its sampled tips, while the internal lineage colorings Y are latent. (KLI Theorem 2 is the *colored* case, Y given, with a filter over x alone; the latent-coloring case here is Theorem 5 with Appendix B.) The camel/human tip types are sample metadata in the sense of KLI Remark 3, entering ϕ multiplicatively through the compatible chop operator, and $L(\theta) = \Pr_\theta\{(T, Z), M\} = \mathbb{E}_\theta[\Pr\{(T, Z), M \mid H_T\}]$. The filter in KLI Theorem 5 is a *weighted* process: with V_t the running importance weight (initial $1/q$, regular boosts, singular rate×boost factors, decay), define $w(t, x, y)\,dx = \mathbb{E}[V_t\,\mathbb{1}_{X_t \in dx,\, Y_t = y}]$, so that $L(\theta) = \sum_{y \in \mathcal{Y}_T(Z)} \int w(T, x, y)\,dx$. It satisfies a filter equation with a driver β, a boost B, and a decay λ.

Driver–boost–decay form of the Theorem 5 filter (KLI Appendix B). At non-event times $t \notin \text{ev}(Z)$,

$$\frac{\partial w}{\partial t}(t, x, y) = \sum_{u \in \mathbb{U}} \sum_{y'} \int w(t, x', y')\,\beta_u^{\text{reg}}(t, x', x, y', y)\,B_u(t, x', x, y', y)\,dx' - \sum_{u \in \mathbb{U}} \sum_{y'} \int w(t, x, y)\,\beta_u^{\text{reg}}(t, x, x', y, y')\,dx' - \lambda(t, x, y)\,w(t, x, y),$$

and at an event $t = e \in \text{ev}(Z)$,

$$w(t, x, y) = \sum_{u \in \mathcal{U}_e} \sum_{y'} \int w(t^-, x', y')\,\beta_{e,u}^{\text{ev}}(t, x', x, y', y)\,B_u(t, x', x, y', y)\,dx',$$

where $\mathcal{U}_e$ are the marks compatible with event e and the post-event coloring y (a typed leaf: one mark; a fork with both post-event branches C: $\{T_{CC}\}$; both H: $\{T_{HH}\}$; a mixed fork: $\{T_{HC}, T_{CH}\}$) and $\beta_{e,u}^{\text{ev}} = \alpha_u\,\pi_u$ is supported on $\text{ev}(Z)$. The sum runs over the full mark collection $\mathbb{U}$; the singular sampling marks have $\beta_u^{\text{reg}} = 0$ on open intervals and so contribute to the regular part only through the decay λ, not the flow. Since $\sum_{y'} \pi_u = 1$, the outflow collapses to $-w(t, x, y) \sum_u \int \alpha_u^{\text{reg}}(t, x, x'; y)\,dx'$ (the reduced removal rate depends on y through the threshold $\mathbb{1}_{I_d > \ell_d(y)}$). The likelihood is $L(\theta) = \sum_{y \in \mathcal{Y}_T(Z)} \int w(T, x, y)\,dx$, the sum over colorings of Z consistent with the typed tips.

Algorithm B1 (SMC). Because w has no closed form in general, KLI give a *simple sequential importance-sampling* algorithm (no resampling in the displayed Algorithm B1; resampling is an optional improvement): P particles $\{(x^{(p)}, y^{(p)})\}$ are propagated forward, their weights $V^{(p)}$ multiplied by the boost B at each regular jump and by the rate×boost $A\,B$ at each $t_i \in \text{ev}(Z)$, and decayed by $\int \lambda \, dt$ between events. The estimator $\hat{L}(\theta) = \frac{1}{P}\sum_p V_T^{(p)}$ is unbiased for $L(\theta)$ (KLI Lemma 3); $\log \hat{L}$ is then a (biased) estimate of $\log L$, $\mathbb{E}[\log \hat{L}] \le \log L$ by Jensen. These estimates support likelihood profiling and MCAP; an IF2 fit would embed the same model-specific filter components in the usual resampling and parameter-perturbation recursion.

3 Model Specification

Demes $\mathcal{D} = \{C, H\}$ (I_C: infectious camels; I_H: infectious humans). State $x = (S_C, I_C, S_H, I_H)$, deme-occupancy $n(x) = (I_C, I_H)$.

u	Description	Rate α_u	$r^u = (r_C, r_H)$
TCC	camel→camel birth	$\beta_{CC} S_C I_C / N_C$	$(2, 0)$
THH	human→human birth	$\beta_{HH} S_H I_H / N_H$	$(0, 2)$
THC	camel→human birth	$\beta_{HC} S_H I_C / N_C$	$(1, 1)$
TCH	human→camel birth	$\beta_{CH} S_C I_H / N_H$	$(1, 1)$
RC	camel removal (death)	$\gamma_C I_C$	$(0, 0)$
RH	human removal (death)	$\gamma_H I_H$	$(0, 0)$
SC	camel sample/death‡	$\chi_C I_C$	$(0, 0)$
SH	human sample/death‡	$\chi_H I_H$	$(0, 0)$
BC,BH,DC,DH	demography (neutral)	(see YAML)	$(0, 0)$

‡SC and SH are compound sample/death-type events (KLI Fig. 4G): each simultaneously collects a sample (leaf node inserted) and removes the host. In the phylopomp YAML these appear as the jump mark `sample_death`. The cross-deme births THC and TCH are the distinguishing events: each has one slot in C and one in H, with the new child born from a parent in the opposite deme.

The generic `TwoSpecies` implementation distinguishes a per-capita sampling rate ψ_d from a culling probability c_d; the MERS specification fixes $c_C = c_H = 1$, so every sampled infectious host is culled and we write $\chi_d = \psi_d$. A model with $c_d < 1$ would require separate non-destructive sample and sample/death marks.

4 Coupling Between Camel and Human Lineages

In the two-strain companion (Note S), no production vector had nonzero components in both strains, so the genealogy decoupled. For MERS that fails: THC has $r = (1, 1)$ with parent in C and child born into H; TCH has $r = (1, 1)$ with parent in H and child born into C. Both span both demes and produce cross-deme branch points $\kappa^{bb'}_{CHC}$ (THC) and $\kappa^{bb'}_{HCH}$ (TCH). Consequently a lineage's deme is not a time-invariant label (a human lineage traces back to a camel at a THC event), the operators $\sigma^b_{CH}, \sigma^b_{HC}$ are genuine cross-deme operators, and the coloring y couples both demes.

Remark 4.1. *A cross-deme branch point $\kappa^{bb'}_{CHC}$ is the local fork structure of camel-to-human spillover: in a* fully colored *genealogy, going backward, a human lineage b and a camel lineage b' share a common ancestor in the camel deme. In the* obscured *genealogy it is one latent explanation of a mixed internal fork, summed over by the filter (the parent deme, the newly infected host, and the T_{HC}-vs-T_{CH} direction are not directly observed). This is the structure visible once a coloring is imposed on the Dudas et al. MERS phylogeny.*

Coloring convention. Throughout the MERS-specific derivation, y denotes only the *deme coloring* of the active branches, obtained from KLI's full coloring $Y = (d, m)$ by summing out the inline event-indicator component m. On this reduced state a same-deme swap is the identity, $\sigma^b_{CC} y = y$ and $\sigma^b_{HH} y = y$, even though the corresponding operator is not an identity on KLI's full coloring (it changes the event indicator).

For a transition of the reduced color-only process, let $\Phi_u(y^-, y^+)$ denote the sum of the KLI compatibility factors ϕ_u over all full-coloring transitions that collapse to the same reduced transition. The proposal kernel π_u is defined on reduced colorings, and the boost is therefore

$$B_u(y^-, y^+) = \frac{\Phi_u(y^-, y^+)}{\pi_u(y^-, y^+)}.$$

When a reduced transition corresponds to a single full-coloring transition, $\Phi_u = \phi_u$. When several collapse under the reduction, Φ_u is their sum, evaluated in closed form by the Chu–Vandermonde identity. Thus $\beta_u B_u = \alpha_u \pi_u \cdot (\Phi_u/\pi_u) = \alpha_u \Phi_u$.

5 Construction of Compatibility Terms

Each mark u contributes the binomial ratio ϕ_u and the indicator $Q_u(y, y')$, derived in four steps.

5.1 Step A: production slots and ancestral demes

For each slot, the *ancestral deme* is the deme the lineage came from immediately before the event: a continuing-parent slot or a same-deme new child has ancestral deme d; a new child in a different deme d' has ancestral deme d (the parent's deme), not d'.

5.2 Step B: enumerate saturations

For each deme d, $s_d \in \{0, 1, \ldots, \min(r_d, \ell_d)\}$; feasibility also requires $I_d \geq \max\{\ell_d, r_d\}$ (automatic post-event for the production marks). For the *transmission* marks, whose total production is two, full saturation $s_C = r_C$ and $s_H = r_H$ is a branch point, handled only at data-event times; for the $r = (0, 0)$ marks (removal, sample/death, neutral) full saturation $s = r = (0, 0)$ involves no fork.

5.3 Step C: compute the binomial ratio

$$\phi_u = \prod_{d \in \{C,H\}} \frac{\binom{I_d - \ell_d}{r_d - s_d}}{\binom{I_d}{r_d}},$$

where I_d is the *post-event* occupancy and $\ell_d = |\operatorname{col}_d(Y_t)|$ is the *post-event* pruned count (the count attached to the destination coloring in the filter). Demes with $r_d = 0$ contribute factor 1. The $\binom{\ell_d}{s_d}$ ways to choose which pruned lineages fill pruned slots are handled by the sum over colorings in the filter equations.

5.4 Step D: the coloring operator

Single slot occupied by pruned b (ancestral deme d_{pre}, post-event deme d_{post}): $y' = \sigma^b_{d_{\text{pre}}, d_{\text{post}}} y$. On the reduced color-only state this is the identity when $d_{\text{pre}} = d_{\text{post}}$; on KLI's full coloring it is still a distinct inline-event transition because the event-indicator component changes. Two slots occupied (branch point), pruned b, b' with common ancestor deme d_{anc}: $y' = \kappa^{bb'}_{d_{\text{anc}},d_b,d_{b'}} y$. No slot occupied: $y' = y$.

6 Event-Specific Derivations

Throughout, I_C, I_H are post-event occupancies and ℓ_C, ℓ_H are the pruned counts in the post-event coloring y.

6.1 TCC: within-camel birth, $r = (2, 0)$

Both slots (parent C1, new child C2) have ancestral deme C. With $s_C \in \{0, 1, 2\}$, $s_H = 0$:

$$s_C{=}0: \ \frac{(I_C - \ell_C)(I_C - \ell_C - 1)}{I_C(I_C - 1)}, \quad s_C{=}1: \ \frac{2(I_C - \ell_C)}{I_C(I_C - 1)}, \quad s_C{=}2: \ \frac{2}{I_C(I_C - 1)}.$$

Operators: $s_C{=}0$: $y' = y$; $s_C{=}1$: $\sigma^b_{CC} y = y$ (identity); $s_C{=}2$: $y' = \kappa^{bb'}_{CCC} y$ (within-camel coalescence).

6.2 THH: within-human birth, $r = (0, 2)$

Exactly symmetric with $C \leftrightarrow H$: ratios $\frac{(I_H - \ell_H)(I_H - \ell_H - 1)}{I_H(I_H - 1)}$, $\frac{2(I_H - \ell_H)}{I_H(I_H - 1)}$, $\frac{2}{I_H(I_H - 1)}$; operators y, $\sigma^b_{HH} y = y$, $\kappa^{bb'}_{HHH} y$.

6.3 THC: camel-to-human spillover, $r = (1, 1)$

C-slot is the continuing camel parent (ancestral deme C); H-slot is the new human child, ancestral deme C (not H). With $(s_C, s_H) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\}$:

$$(0, 0): \frac{(I_C - \ell_C)(I_H - \ell_H)}{I_C I_H}, \quad (0, 1): \frac{I_C - \ell_C}{I_C I_H}, \quad (1, 0): \frac{I_H - \ell_H}{I_C I_H}, \quad (1, 1): \frac{1}{I_C I_H}.$$

Operators: $(0, 0)$: $y' = y$; $(0, 1)$: $y' = \sigma^b_{CH} y$, $b \in \text{col}_C(y)$ pre-event (the new human child traces to a camel parent); $(1, 0)$: $y' = \sigma^b_{CC} y = y$ (continuing parent); $(1, 1)$: $y' = \kappa^{bb'}_{CHC} y$ ($d_{\text{anc}} = C$, $d_b = H$, $d_{b'} = C$) — the spillover branch point.

6.4 TCH: human-to-camel spillover, $r = (1, 1)$

Exact mirror with $C \leftrightarrow H$; the slot roles swap (in TCH the C-slot is the child):

$$(0, 0): \frac{(I_H - \ell_H)(I_C - \ell_C)}{I_H I_C}, \quad (1, 0): \frac{I_H - \ell_H}{I_H I_C}, \quad (0, 1): \frac{I_C - \ell_C}{I_H I_C}, \quad (1, 1): \frac{1}{I_H I_C}.$$

Operators: $(1, 0)$: $y' = \sigma^b_{HC} y$, $b \in \text{col}_H(y)$ (new camel child from a human parent); $(0, 1)$: $\sigma^b_{HH} y = y$; $(1, 1)$: $y' = \kappa^{bb'}_{HCH} y$.

6.5 RC and RH: removal (death)

$r = (0, 0)$, $s = (0, 0)$ forced, $\phi = 1$, $y' = y$, with post-event support $\mathbb{1}_{I_d(x) \geq \ell_d(y)}$ for $d \in \{C, H\}$. As a transition *into* the post-event state x, removal is compatible whenever $I_d \geq \ell_d$ (the removed host was untracked). The *strict* $I_d > \ell_d$ is the driver-*outflow* condition: removal from the current state needs $I_d - 1 \geq \ell_d$ at the destination. When $I_d = \ell_d$ the outflow has no compatible target and becomes the sub-threshold decay $\gamma_d I_d \mathbb{1}_{I_d \leq \ell_d}$ in λ (§8).

6.6 SC and SH: destructive sampling (sample/death)

The phylopomp mark `sample_death(camel)` is compound: it inserts a leaf and removes the host from I_C. Its genealogical role splits by time:

- **Observed leaf** ($t \in S_0^{(C)}$, a data-event time): the sampled camel is a leaf; singular update.
- **Sample-type jump at** $t \notin \mathrm{ev}(Z)$: not in the data tree; contributes to λ.

Steps A–B. $r = (0,0)$: no production slots (the host is consumed, not born into a slot); $s = (0,0)$ forced by $s_d \le r_d$.

Step C. Since $r = (0,0)$, KLI Eq. (9) gives the empty-product value $\phi_{\mathrm{SC}} = 1$, as it does for the removal event R_C. Hence the singular contribution is carried by the event rate $\alpha_{\mathrm{SC}}(t, x', x) = \chi_C I_C(x')$, where $I_C(x') = I_C(x) + 1$ because x' is the state immediately before the sampled host is removed. No additional factor $1/I_C$ appears. For comparison, when a mark has $r = 1$, factors such as $(I - \ell)/I$ or $1/I$ may arise directly from KLI's binomial ratio; here $r = 0$, so that ratio is 1.

Step D. Data-event time: leaf a sampled ($a \in \mathrm{col}_C(y)$), $y' = \chi_C^{a\emptyset} y$. Non-data time: contributes to λ only. The full camel sampling rate $\chi_C I_C$ enters λ because the exact likelihood conditions on no sample at $t \notin \mathrm{ev}(Z)$. SH is symmetric.

7 Complete Reference Table

The first table is the *uncollapsed* KLI compatibility table. In this table y and y' denote KLI's full pre-event and post-event colorings, including the inline event-indicator component m. Thus same-deme swap rows such as $y' = \sigma_{CC}^{b} y$ are distinct from $y' = y$ in this full table. The quantities I_C, I_H and $\ell_C = \ell_C(y')$, $\ell_H = \ell_H(y')$ are post-event quantities, while $\mathrm{col}_C(y)$ and $\mathrm{col}_H(y)$ refer to the pre-event coloring.

u	r	(s_C, s_H)	$Q_u(y, y')$: compatible transition	ratio ϕ_u [†]
TCC	$(2,0)$	$(0,0)$	$y' = y$	$\dfrac{(I_C-\ell_C)(I_C-\ell_C-1)}{I_C(I_C-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(1,0)$	$y' = \sigma^b_{CC} y$, $\exists b \in \mathrm{col}_C(y)$	$\dfrac{2(I_C-\ell_C)}{I_C(I_C-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(2,0)$	$y' = \kappa^{bb'}_{CCC} y$, $b \cup b' \in \mathrm{col}_C(y)$	$\dfrac{2}{I_C(I_C-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
THH	$(0,2)$	$(0,0)$	$y' = y$	$\dfrac{(I_H-\ell_H)(I_H-\ell_H-1)}{I_H(I_H-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(0,1)$	$y' = \sigma^b_{HH} y$, $\exists b \in \mathrm{col}_H(y)$	$\dfrac{2(I_H-\ell_H)}{I_H(I_H-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(0,2)$	$y' = \kappa^{bb'}_{HHH} y$, $b \cup b' \in \mathrm{col}_H(y)$	$\dfrac{2}{I_H(I_H-1)}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
THC	$(1,1)$	$(0,0)$	$y' = y$	$\dfrac{(I_C-\ell_C)(I_H-\ell_H)}{I_C I_H}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(0,1)$	$y' = \sigma^b_{CH} y$, $\exists b \in \mathrm{col}_C(y)$	$\dfrac{I_C-\ell_C}{I_C I_H}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(1,0)$	$y' = \sigma^b_{CC} y$, $\exists b \in \mathrm{col}_C(y)$	$\dfrac{I_H-\ell_H}{I_C I_H}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(1,1)$	$y' = \kappa^{bb'}_{CHC} y$, $b \cup b' \in \mathrm{col}_C(y)$	$\dfrac{1}{I_C I_H}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
TCH	$(1,1)$	$(0,0)$	$y' = y$	$\dfrac{(I_H-\ell_H)(I_C-\ell_C)}{I_H I_C}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(1,0)$	$y' = \sigma^b_{HC} y$, $\exists b \in \mathrm{col}_H(y)$	$\dfrac{I_H-\ell_H}{I_H I_C}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(0,1)$	$y' = \sigma^b_{HH} y$, $\exists b \in \mathrm{col}_H(y)$	$\dfrac{I_C-\ell_C}{I_H I_C}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
		$(1,1)$	$y' = \kappa^{bb'}_{HCH} y$, $b \cup b' \in \mathrm{col}_H(y)$	$\dfrac{1}{I_H I_C}\,\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
RC	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
RH	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
SC	$(0,0)$	$(0,0)$	$y' = \chi^{a\emptyset}_C y$, $a \in \mathrm{col}_C(y)$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$ $(\alpha_{\mathrm{SC}} = \chi_C I_C(x'))$
SH	$(0,0)$	$(0,0)$	$y' = \chi^{a\emptyset}_H y$, $a \in \mathrm{col}_H(y)$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$ $(\alpha_{\mathrm{SH}} = \chi_H I_H(x'))$
BC	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
BH	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
DC	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$
DH	$(0,0)$	$(0,0)$	$y' = y$	$\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$

[†]For every row the displayed weight is the production-slot binomial ratio $\phi_u = \prod_d \binom{I_d-\ell_d}{r_d-s_d}/\binom{I_d}{r_d}$ with I_d, ℓ_d post-event, times the support indicators $\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$ (KLI Eq. 37): the ratio is defined only where each deme has at least as many infectious hosts as pruned lineages. For RC, RH, SC, SH, BC/BH/DC/DH: all have $r = (0,0)$, the empty product, so the entry is $\mathbb{1}_{I_C\ge\ell_C}\,\mathbb{1}_{I_H\ge\ell_H}$ itself. For SC and SH, $\phi_u = 1$ and the singular update is carried by the full rate $\alpha_{S_d} = \chi_d I_d(x')$ with $I_d(x') = I_d(x) + 1$, on compatible chop transitions $\chi^{a\emptyset}_d$. RC/RH compatibility is the post-event support $\mathbb{1}_{I_d\ge\ell_d}$ (the removed host was untracked); the strict $I_d > \ell_d$ is *not* a compatibility condition but the driver-outflow condition (§5.5, §8), since the outflow destination $I_d - 1$ must still satisfy $\ge \ell_d$.

Reading the uncollapsed table. Each row is one compatible transition; all unlisted (y, y') have $Q_u = 0$. Conditions $\exists b \in \mathrm{col}_d(y)$ refer to the pre-event coloring (Theorem 1: $Q_u(Y_{t^-}, Y_t)$). The binomial ratio uses post-event occupancies I_C, I_H *and* post-event pruned counts $\ell_C(Y_t), \ell_H(Y_t)$; only Q_u uses the pre-event coloring. The cross-deme forks THC/TCH $(1,1)$ have $b \cup b' \in \mathrm{col}_C(y)$ (resp. col_H) pre-event, with post-event demes encoded in the κ subscript. SC/SH have $r = (0,0)$, $s = (0,0)$, $\phi = 1$, with singular rate $\alpha_{S_d} = \chi_d I_d(x')$.

Reduction over the event indicator. Following KLI §5.3, write the full coloring as $Y = (d, m)$, where d is the deme coloring and m the event-indicator component, and define

$$w(t, x, d) = \sum_m w(t, x, d, m).$$

Transitions that differ only in m but induce the same reduced deme-coloring transition $d \to d'$ therefore combine; for each mark u, $\Phi_u(d, d')$ denotes the resulting summed compatibility factor. The reduced table below lists these Φ_u. They are *not* in general the proposal boosts: for a reduced proposal kernel π_u, the boost is $B_u = \Phi_u / \pi_u$ (§9).

u	color change	reduced factor Φ_u
TCC	$d' = d$	$\left(1 - \frac{\ell_C(\ell_C-1)}{I_C(I_C-1)}\right) \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \kappa^{bb'}_{CCC} d$	$\frac{2}{I_C(I_C-1)} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
THH	$d' = d$	$\left(1 - \frac{\ell_H(\ell_H-1)}{I_H(I_H-1)}\right) \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \kappa^{bb'}_{HHH} d$	$\frac{2}{I_H(I_H-1)} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
THC	$d' = d$	$\left(1 - \frac{\ell_H}{I_H}\right) \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \sigma^{b}_{CH} d$	$\dfrac{I_C - \ell_C}{I_C I_H} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \kappa^{bb'}_{CHC} d$	$\dfrac{1}{I_C I_H} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
TCH	$d' = d$	$\left(1 - \frac{\ell_C}{I_C}\right) \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \sigma^{b}_{HC} d$	$\dfrac{I_H - \ell_H}{I_C I_H} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
	$d' = \kappa^{bb'}_{HCH} d$	$\dfrac{1}{I_C I_H} \mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
SC	$d' = \chi^{a\emptyset}_{C} d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
SH	$d' = \chi^{a\emptyset}_{H} d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
RC	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
RH	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
BC	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
BH	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
DC	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$
DH	$d' = d$	$\mathbb{1}_{I_C \ge \ell_C} \mathbb{1}_{I_H \ge \ell_H}$

8 Genealogy-Specific Filter Terms

There are twelve genealogy-specific terms: six singular update terms (chop χ and fork κ rows, weight jumps at $t \in \mathrm{ev}(Z)$) and six regular inflow components obtained from the non-fork rows after summing over the event-indicator component, which the regular filter equation (Eq. 13) assembles.

Convention. In the singular equations y is the post-event filter coloring; reversal operators (χ^r, κ^r) recover the pre-event coloring, and $\widetilde{w}(t, x', y') := \lim_{s \uparrow t} w(s, x', y') = w(t^-, x', y')$ is the *left-limit* weight at the source state x' and the recovered pre-event coloring y' (KLI's left-limit notation, p. 29; the reversals are the operators $\chi^r, \sigma^r, \kappa^r$, p. 20; $\widetilde{w}$ is not a separate "reversal weight"). In the regular equations y is the current (post-event) filter state. KLI Eq. 9 gives the post-event support condition $\mathbb{1}_{I_C(x) \ge \ell_C(y)} \mathbb{1}_{I_H(x) \ge \ell_H(y)}$; throughout, $w \equiv 0$ outside $\mathcal{S} = \{I_C \ge \ell_C,\ I_H \ge \ell_H\}$, and we display the support indicators on the singular equations and take the regular components I_7–I_{12} to hold on $\mathcal{S}$ (without repeating the factor). *Eqs. 5–6 are the two components of the mixed-fork* *singular update*: the assembled update at $t \in \mathrm{ev}(Z)$ sums over the feasible marks and predecessor colorings. A typed leaf collapses to one term (the tip deme C/H is observed); a mixed internal fork does not; it sums over THC and TCH (camel parent/human child vs. human parent/camel child), since internal demes are latent.

8.1 Equations 1–6: Singular

Eq. 1 — Camel leaf $t \in S_0^{(C)}$ (chop $\chi_C^{a\emptyset}$):

$$S_{\mathrm{SC}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\Big(t,x',\chi^r_{a\emptyset,C}y\Big)\,\alpha_{\mathrm{SC}}(t,x',x)\,dx', \tag{1}$$

with $\alpha_{\mathrm{SC}} = \chi_C I_C(x') = \chi_C(I_C(x)+1)$.

Eq. 2 — Human leaf $t \in S_0^{(H)}$ (chop $\chi_H^{a\emptyset}$):

$$S_{\mathrm{SH}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\Big(t,x',\chi^r_{a\emptyset,H}y\Big)\,\alpha_{\mathrm{SH}}(t,x',x)\,dx', \tag{2}$$

with $\alpha_{\mathrm{SH}} = \chi_H I_H(x') = \chi_H(I_H(x)+1)$.

Remark 8.1 (Why $\phi_{S_d}=1$)**.** *For S_d the production vector is $r=(0,0)$, so KLI Eq. (9) gives $\phi_{S_d}=1$. The singular update is therefore weighted by the full event rate $\alpha_{S_d}(t,x',x)=\chi_d I_d(x')$, together with the compatibility condition and the post-event support indicator. Since $I_d(x')=I_d(x)+1$, the pre-removal occupancy enters through α_{S_d}; there is no additional $1/I_d$ factor.*

Eq. 3 — Within-camel fork $t \in B(CC)$ (fork $\kappa_{CCC}^{bb'}$):

$$S_{\mathrm{TCC}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\big(t,x',\kappa^r_{bb',CCC}y\big)\,\alpha_{\mathrm{TCC}}\,\frac{2}{I_C(I_C-1)}\,dx'. \tag{3}$$

The factor 2 absorbs both label assignments of b, b' to the two C-slots.

Eq. 4 — Within-human fork $t \in B(HH)$ (fork $\kappa_{HHH}^{bb'}$):

$$S_{\mathrm{THH}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\big(t,x',\kappa^r_{bb',HHH}y\big)\,\alpha_{\mathrm{THH}}\,\frac{2}{I_H(I_H-1)}\,dx'. \tag{4}$$

Eq. 5 — Cross-deme fork THC component $t \in B(CH)$ (fork $\kappa_{CHC}^{bb'}$):

$$S_{\mathrm{THC}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\big(t,x',\kappa^r_{bb',CHC}y\big)\,\alpha_{\mathrm{THC}}\,\frac{1}{I_C I_H}\,dx'. \tag{5}$$

In the post-event y: $b \in \mathrm{col}_H(y)$ (new H-child), $b' \in \mathrm{col}_C(y)$ (continuing C-parent).

Eq. 6 — Cross-deme fork TCH component $t \in B(HC)$ (fork $\kappa_{HCH}^{bb'}$):

$$S_{\mathrm{TCH}}(t,x,y) = \mathbb{1}_{I_C(x)\ge \ell_C(y)}\, \mathbb{1}_{I_H(x)\ge \ell_H(y)} \int \widetilde{w}\big(t,x',\kappa^r_{bb',HCH}y\big)\,\alpha_{\mathrm{TCH}}\,\frac{1}{I_H I_C}\,dx'. \tag{6}$$

In Eq. 6 the labels are oriented for the TCH explanation: $b \in \mathrm{col}_C(y)$ is the new camel child and $b' \in \mathrm{col}_H(y)$ the continuing human parent (the swapped branch order relative to Eq. 5, matching the $\kappa^r_{bb',HCH}$ subscript $d_b = C$, $d_{b'} = H$). At a mixed internal fork both THC and TCH are latent explanations of the same obscured branch point (the parent deme is unobserved), so S_{THC} and S_{TCH} are two *component* terms entering the same singular update; each child orientation (C,H) or (H,C) carries binomial ratio $1/(I_C I_H)$, and the forward fork proposal (§9) splits $\frac{1}{2}+\frac{1}{2}$ between them.

8.2 Assembled singular filter

Write $S_{\mathrm{SC}}, S_{\mathrm{SH}}, S_{\mathrm{TCC}}, S_{\mathrm{THH}}, S_{\mathrm{THC}}, S_{\mathrm{TCH}}$ for the six component terms above (Eqs. 1–6). At a data-event time $t = e \in \mathrm{ev}(Z)$ the singular filter is their compatible sum — the event-time analogue of the regular $\sum_{k=7}^{12} I_k$ inflow (§7.4):

$$w(t,x,y) = \sum_{u \in \mathcal{U}_e(y)} S_u(t,x,y), \qquad t = e \in \mathrm{ev}(Z),$$

where $\mathcal{U}_e(y)$ is the set of marks compatible with the observed event e *and* the post-event coloring y:

$$\mathcal{U}_e(y) = \begin{cases} \{\mathrm{SC}\}, & \text{camel leaf,} \\ \{\mathrm{SH}\}, & \text{human leaf,} \\ \{\mathrm{TCC}\}, & \text{fork with two } C \text{ children,} \\ \{\mathrm{THH}\}, & \text{fork with two } H \text{ children,} \\ \{\mathrm{THC}, \mathrm{TCH}\}, & \text{mixed fork.} \end{cases}$$

Equivalently, the analytic (fixed-destination) singular update is

$$w(t,x,y) = \begin{cases} S_{\mathrm{SC}}(t,x,y), & \text{camel leaf,} \\ S_{\mathrm{SH}}(t,x,y), & \text{human leaf,} \\ S_{\mathrm{TCC}}(t,x,y), & y \text{ has two } C \text{ children,} \\ S_{\mathrm{THH}}(t,x,y), & y \text{ has two } H \text{ children,} \\ S_{\mathrm{THC}}(t,x,y) + S_{\mathrm{TCH}}(t,x,y), & y \text{ is mixed.} \end{cases}$$

A typed leaf or a same-deme post-event coloring fixes a unique compatible mark, so the sum has one term; only a mixed fork sums two components (the parent deme is latent). A same-deme mark contributes a single coloring ($\pi = 1$, the factor 2 carried in ϕ: $B_{\mathrm{TCC}} = 2/[I_C(I_C-1)]$); each cross-deme component contributes two child orientations ($\pi = \frac{1}{2}$ each, $B_{\mathrm{THC}} = \phi_{\mathrm{THC}}/\frac{1}{2} = 2/(I_C I_H)$). The forward, parent-conditioned (SMC) enumeration of these same transitions — which a particle realizes by conditioning on the carried parent deme — is deferred to §9 and §11, to keep this analytic update impossible to misread.

8.3 Equations 7–12: Regular

Chu–Vandermonde simplification. After summing over the event-indicator component, any same-deme non-fork single-slot transition collapses to the same reduced coloring y (the single-slot swap satisfies $\sigma^b_{dd} y = y$). In the cross-deme birth marks, the continuing-parent slot contributes to the identity-coloring term, while the new-child slot crossing demes produces a cross-coloring term.

Eq. 7 — TCC ($\sigma^b_{CC} = \mathrm{id}$; C-V):

$$I_7(t,x,y) = \int w(t,x',y)\, \alpha_{\mathrm{TCC}}(t,x',x) \left[1 - \frac{\ell_C(\ell_C-1)}{I_C(I_C-1)}\right] dx'. \tag{7}$$

The bracketed factor is the reduced non-fork contribution for TCC. The missing mass corresponds to unobserved within-camel branch points and is accounted for through the regular birth inflow/outflow balance (§8), not through λ; the same holds for THH (Eq. 8).

Eq. 8 — THH ($\sigma^b_{HH} = \mathrm{id}$; C-V):

$$I_8(t,x,y) = \int w(t,x',y)\, \alpha_{\mathrm{THH}}(t,x',x) \left[1 - \frac{\ell_H(\ell_H-1)}{I_H(I_H-1)}\right] dx'. \tag{8}$$

Eq. 9 — THC C-V term (C-slot stays, $\sigma^b_{CC} = \mathrm{id}$):

$$I_9(t,x,y) = \int w(t,x',y)\,\alpha_{\mathrm{THC}}(t,x',x)\,\frac{I_H - \ell_H}{I_H}\,dx'. \tag{9}$$

Eq. 10 — THC cross-coloring (H-slot enters; swap σ^b_{CH}):

$$I_{10}(t,x,y) = \sum_{b\in\mathrm{col}_H(y)} \int w\big(t,x',\sigma^r_{b,CH}y\big)\,\alpha_{\mathrm{THC}}(t,x',x)\,\frac{I_C - \ell_C(y)}{I_C I_H}\,dx'. \tag{10}$$

The coefficient uses the post-event (destination) count $\ell_C(y)$: in y, $b \in \mathrm{col}_H(y)$, so b is not counted in $\ell_C(y)$. Per KLI Eq. (9) the ratio uses Y_t; cf. SEIRS Eq. (43).

Eq. 11 — TCH C-V term (H-slot stays, $\sigma^b_{HH} = \mathrm{id}$):

$$I_{11}(t,x,y) = \int w(t,x',y)\,\alpha_{\mathrm{TCH}}(t,x',x)\,\frac{I_C - \ell_C}{I_C}\,dx'. \tag{11}$$

Eq. 12 — TCH cross-coloring (C-slot enters; swap σ^b_{HC}):

$$I_{12}(t,x,y) = \sum_{b\in\mathrm{col}_C(y)} \int w\big(t,x',\sigma^r_{b,HC}y\big)\,\alpha_{\mathrm{TCH}}(t,x',x)\,\frac{I_H - \ell_H(y)}{I_H I_C}\,dx'. \tag{12}$$

8.4 Assembled regular filter

Between events ($t \notin \mathrm{ev}(Z)$) the regular filter equation is the sum, over all regular marks, of each mark's inflow integral minus its outflow (KLI Eqs. (35)–(36)). The birth inflows are the integral components $I_7, \ldots, I_{12}$ of §7.3; the removal and demography kernels are point masses, shown here with the transition evaluated (the $w(t, x{\pm}e, y)$ shifted-state form of KLI Eq. (36)). Each birth mark's outflow is its full hazard $\alpha_u(t,x)\,w$. Removal and demography contribute matched gain–loss pairs, and the sampling hazard together with the sub-threshold removal appears as the explicit decay:

$$\begin{aligned}
\frac{\partial w}{\partial t}(t,x,y) = {}& \sum_{k=7}^{12} I_k - \big(\alpha_{\mathrm{TCC}} + \alpha_{\mathrm{THH}} + \alpha_{\mathrm{THC}} + \alpha_{\mathrm{TCH}}\big)\,w(t,x,y) \\
& + \gamma_C(I_C{+}1)\,\mathbb{1}_{I_C \ge \ell_C(y)}\,w(t,x{+}e_C,y) - \gamma_C I_C\,\mathbb{1}_{I_C > \ell_C(y)}\,w(t,x,y) \\
& + \gamma_H(I_H{+}1)\,\mathbb{1}_{I_H \ge \ell_H(y)}\,w(t,x{+}e_H,y) - \gamma_H I_H\,\mathbb{1}_{I_H > \ell_H(y)}\,w(t,x,y) \\
& + B_C\,\mathbb{1}_{S_C \ge 1}\,w(t,x{-}e_{S_C},y) - B_C\,w(t,x,y) \\
& + B_H\,\mathbb{1}_{S_H \ge 1}\,w(t,x{-}e_{S_H},y) - B_H\,w(t,x,y) \\
& + B_C\,w(t,x{+}e_{S_C},y) - B_C\,\mathbb{1}_{S_C>0}\,w(t,x,y) \\
& + B_H\,w(t,x{+}e_{S_H},y) - B_H\,\mathbb{1}_{S_H>0}\,w(t,x,y) \\
& - \big(\chi_C I_C + \chi_H I_H\big)\,w(t,x,y) - \big(\gamma_C I_C\,\mathbb{1}_{I_C \le \ell_C(y)} + \gamma_H I_H\,\mathbb{1}_{I_H \le \ell_H(y)}\big)\,w(t,x,y).
\end{aligned} \tag{13}$$

The births' full mass-removal outflow $\sum_u \alpha_u\,w$ collapses against the birth inflows I_7–I_{12} (Chu–Vandermonde, §7.3) to the reduced driver of §8. Each removal hazard $\gamma_d I_d$ splits into an above-threshold outflow ($I_d > \ell_d(y)$, a compatible transition) and a sub-threshold remainder ($I_d \le \ell_d(y)$); the remainder joins the sampling hazard in the final (decay) line. This $\ge$ versus $>$ split is the removal-compatibility point of §5.5 (cf. KLI Eq. (34)): the inflow to x requires only $I_d \ge \ell_d$, whereas the outflow from x needs the destination $I_d - 1 \geq \ell_d$. Camel death D_C has the *per-capita* rate $B_C S_C / N_C$ (v28; see the changelog), which is already zero exactly when $S_C = 0$, so no separate guard indicator is needed — unlike v25's constant-rate form, an attempted death at $S_C = 0$ never has positive hazard in the first place, and the matching inflow to x (from $x+e_{S_C}$) is likewise unconditional. Human death D_H is the exact mirror: per-capita rate $B_H S_H / N_H$. Births remain the constant rates B_C, B_H (unchanged from v25). This *does* agree with the R `phylopomp` reference model: its `yaml/mers.yml` was changed to the same per-capita form, and likewise dropped its `if (S>0)` guard, in `f636a7e` ("fix MERS demography", 2026-08-01). The sampling marks SC, SH are singular and contribute nothing on open intervals; their full hazards $\chi_C I_C + \chi_H I_H$ enter only as the decay above.

9 Filter Components

9.1 Decay λ

$$\lambda(t,x,y) = \underbrace{\chi_C I_C + \chi_H I_H}_{\text{sampling hazard}} + \underbrace{\gamma_C I_C \, \mathbb{1}_{I_C \leq \ell_C} + \gamma_H I_H \, \mathbb{1}_{I_H \leq \ell_H}}_{\text{sub-threshold removal}}. \tag{14}$$

Sampling hazard. The exact likelihood conditions on no sample-type jump outside $\mathrm{ev}(Z)$; the *full* rate $\chi_C I_C + \chi_H I_H$ enters λ. **Sub-threshold removal.** When $I_C = \ell_C$ any RC would push I_C below ℓ_C; the weight is killed at rate $\gamma_C I_C$ ($\mathbb{1}_{I_C \leq \ell_C} \equiv \mathbb{1}_{I_C = \ell_C}$). **No separate branch-point hazard.** The within-deme unseen branch-point loss arises automatically from Eqs. 7–8 inflow against the TCC/THH share of the (full-birth) outflow; the cross-deme loss likewise from Eqs. 9–12 against the THC/TCH share. Adding either to λ would double-count.

9.2 Assembled filter equation and the driver β

Per-mark driver and reduced exit rate. The per-mark driver is KLI's $\beta_u^{\text{reg}} = \alpha_u^{\text{reg}} \, \pi_u$ (rate times proposal), so that $\beta_u^{\text{reg}} B_u = \alpha_u^{\text{reg}} \, \Phi_u$. Summed over destinations the outflow $\sum_u \sum_{y'} \int w \, \beta_u^{\text{reg}} \, dx'$ collapses (since $\sum_{y'} \pi_u = 1$) to $w \sum_u \int \alpha_u^{\text{reg}}(t,x,x';y)\, dx'$, with the *reduced regular exit rate*

$$\sum_u \int \alpha_u^{\text{reg}}(t,x,x';y)\, dx' = \underbrace{\alpha_{\text{TCC}} + \alpha_{\text{THH}} + \alpha_{\text{THC}} + \alpha_{\text{TCH}}}_{\text{full births}} + \underbrace{\gamma_C I_C \mathbb{1}_{I_C > \ell_C} + \gamma_H I_H \mathbb{1}_{I_H > \ell_H}}_{\text{above-threshold deaths}} + \underbrace{B_C + B_H + \tfrac{B_C}{N_C} S_C + \tfrac{B_H}{N_H} S_H}_{\text{demography (births constant; deaths per-capita, v28)}},$$

with *no* sampling term. Sampling is singular (Eqs. 1–2) and enters λ at full rate, as does the sub-threshold death. Births are kept full so the branch-point-loss arguments of §8.1 hold unchanged. This is KLI's split (Eqs. 45–47); a literal "full CTMC driver" would subtract $\chi_d I_d$ and the sub-threshold $\gamma_d I_d$ twice.

Regular ($t \notin \mathrm{ev}(Z)$), with the driver $\beta_u^{\mathrm{reg}} = \alpha_u^{\mathrm{reg}}\pi_u$ from the box above:

$$\frac{\partial w}{\partial t}(t,x,y) = \sum_{u\in\mathbb{U}}\sum_{y'}\int w(t,x',y')\,\beta_u^{\mathrm{reg}}(t,x',x,y',y)\,B_u(t,x',x,y',y)\,dx'$$

$$- \,w(t,x,y)\sum_{u\in\mathbb{U}}\int \alpha_u^{\mathrm{reg}}(t,x,x';y)\,dx' - \lambda(t,x,y)\,w(t,x,y).$$

Singular ($t = e \in \mathrm{ev}(Z)$): $w(t,x,y) = \sum_{u\in\mathcal{U}_e}\sum_{y'}\int w(t^-,x',y')\,\beta_{e,u}^{\mathrm{ev}}\,B_u\,dx'$ with $\beta_{e,u}^{\mathrm{ev}} = \alpha_u\pi_u$ supported on $\mathrm{ev}(Z)$ (distinct from β^{reg}, which excludes sampling), in the six explicit forms of §7.1.

10 Proposal Kernel

The proposal is a *forward* kernel: from the pre-event coloring y^- it proposes a post-event y^+, so it must satisfy $\sum_{y^+}\pi_u(t,x^-,x^+,y^-,y^+) = 1$ for each fixed source (x^-,y^-) (KLI Thm. 4). For a cross-deme birth the recolored branch sits in the *parent* deme before the event: at a THC event with only the human child tracked, the tracked branch is the ancestral camel lineage $b \in \mathrm{col}_C(y^-)$; the continuing camel host is the untracked branch. The event recolors b $C \to H$ via σ_{CH}^b. Hence the proposal ranges over the source/parent deme:

$$\pi_{\mathrm{THC}}^{\emptyset} = \frac{I_C - \ell_C(y^-)}{I_C},\quad \pi_{\mathrm{THC}}^{b} = \frac{1}{I_C}\quad \big(b \in \mathrm{col}_C(y^-),\ y^+ = \sigma_{CH}^b y^-\big);$$

$$\pi_{\mathrm{TCH}}^{\emptyset} = \frac{I_H - \ell_H(y^-)}{I_H},\quad \pi_{\mathrm{TCH}}^{b} = \frac{1}{I_H}\quad \big(b \in \mathrm{col}_H(y^-),\ y^+ = \sigma_{HC}^b y^-\big),$$

each summing to 1 over the $\ell_d(y^-)+1$ outgoing destinations. This follows the same source/parent-deme indexing used in KLI's SEIRS transmission proposal ($\pi_{\mathrm{Trans}}^b = 1/I$ over col_I, $\pi_{\mathrm{Trans}}^{\emptyset} = (I-\ell_I)/I$). The boosts $B_u = \Phi_u/\pi_u$ are then *not* generally one:

$$B_{\mathrm{THC}}^{\emptyset} = \frac{(I_H-\ell_H)/I_H}{(I_C-\ell_C(y^-))/I_C} = \frac{I_C(I_H-\ell_H)}{I_H(I_C-\ell_C)},\qquad B_{\mathrm{THC}}^{b} = \frac{(I_C-\ell_C(y^+))/(I_C I_H)}{1/I_C} = \frac{I_C-\ell_C(y^+)}{I_H},$$

with the TCH mirror. For the within-deme births TCC/THH every non-fork regular case leaves y unchanged, so $\pi(y,y) = 1$ and the collapsed C-V factor is carried entirely in B, e.g. $B_{\mathrm{TCC}} = 1 - \dfrac{\ell_C(\ell_C-1)}{I_C(I_C-1)}$. In the analytic filter equation, $\beta B = \alpha_u\Phi_u$ (the π-factors cancel); the proposal matters only for the particle filter.

Singular fork proposals. At an observed branch point the proposal is over the post-event coloring, and the count of distinct colorings (not an auxiliary slot label) determines it. A *same-deme* fork (TCC/THH) gives a *single* coloring (both children take the parent's deme; swapping the two distinguishable subtrees leaves y unchanged), so $\pi = 1$ and the factor 2 stays in ϕ: $B_{\mathrm{TCC}} = \phi_{\mathrm{TCC}} = 2/[I_C(I_C-1)]$ (THH mirror). A *cross-deme* fork (THC/TCH) gives *two* distinct colorings — the two subtrees can be (C,H) or (H,C) — so $\pi = \frac{1}{2}$ per orientation and $B_{\mathrm{THC}} = \phi_{\mathrm{THC}}/\frac{1}{2} = 2/(I_C I_H)$ (TCH mirror). KLI's SEIRS branch-point proposal uses the same $\frac{1}{2}+\frac{1}{2}$ orientation split.

Full-support proposal for the weighted process. The naïve relative-abundance proposal above is not support-safe: when $I_C = \ell_C(y^-)$ (all camels tracked), $\pi^{\emptyset}_{\mathrm{THC}} = 0$, while the collapsed identity-coloring weight $\Phi^{\emptyset}_{\mathrm{THC}} := (I_H - \ell_H)/I_H$ (the aggregate of Eq. 9 over the identity transition) remains positive. Hence $B^{\emptyset} = \Phi^{\emptyset}/\pi^{\emptyset}$ divides by zero. KLI Theorem 4 requires $\pi_u > 0$ whenever $\alpha_u Q_u > 0$; in the reduced process this corresponds to every transition for which $\alpha_u \Phi_u > 0$. The condition is needed not only for simulation but wherever the representation $\beta_u = \alpha_u \pi_u$, $B_u = \Phi_u/\pi_u$ is used, since otherwise $\beta B = 0 \cdot \infty$ at a boundary transition. Mix in a uniform floor over the $\ell_C(y^-) + 1$ outcomes: with $m_C = \ell_C(y^-)$ and $0 < \varepsilon \le 1$,

$$\pi^{\emptyset}_{\mathrm{THC}} = (1-\varepsilon)\frac{I_C - m_C}{I_C} + \frac{\varepsilon}{m_C + 1}, \qquad \pi^{b}_{\mathrm{THC}} = (1-\varepsilon)\frac{1}{I_C} + \frac{\varepsilon}{m_C + 1} \quad (b \in \mathrm{col}_C(y^-)),$$

($\varepsilon = 1$ gives the fully uniform $\pi = 1/(m_C + 1)$); this ε-floored π is one support-safe choice among many; with it $\beta = \alpha\pi$, $B = \Phi/\pi$, and one recomputes B_u from it. The TCH mirror floors over $\mathrm{col}_H(y^-)$. Since $\beta_u B_u = \alpha_u \pi_u \cdot (\Phi_u/\pi_u) = \alpha_u \Phi_u$, the target filter is independent of ε. Changing ε alters the proposal dynamics and the resulting importance weights, but not the analytic filter equation.

Forward vs. inflow indexing. Eq. 10 sums its inflow over $b \in \mathrm{col}_H(y^+)$, the *backward* enumeration of predecessor colorings feeding a fixed destination. The proposal is the *forward* object and is indexed by $\mathrm{col}_C(y^-)$; the two label the same transitions from opposite ends. A destination-indexed proposal $\pi^b = 1/I_H$ over $\mathrm{col}_H(y^+)$ does *not* normalize over y^+ from a fixed y^- (its mass is $1 + (\ell_C(y^-) - \ell_H(y^-))/I_H \neq 1$), so it is not a valid kernel; the source-deme form above is.

10.1 Guided-family particle-filter kernels: soft, guided, and hard

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Terminology. “Naive,” “soft,” “guided,” and “hard” as names for the four proposal kernels, the uniform mixture floor ε, and the auxiliary-intensity symbol κ_ε below are *this repository’s own vocabulary*; none of them appear in KLI’s paper. Every other symbol in this subsection — π_u, Φ_u, B_u, α_u, ℓ_d, I_d, col_d — is KLI’s own notation, used exactly as already defined in §1 and §4; in particular π_ε below is a specific, support-safe choice of the same proposal π_u introduced at the start of this section, not a new kind of object. See the Plain-Language Overview at the front of this document for the intuition behind these four kernels before reading the derivation below.

The proposals above (naïve) never consult the filter guide. This repository’s three guided-family kernels — soft, guided, and hard — instead steer the same proposal π using the guide’s forward-computed present-probabilities (at root and fork proposals) or, in the regular process, its relative hazards r_b (below), while every kernel keeps the same guarantee: $\pi_u > 0$ whenever $\alpha_u \Phi_u > 0$, so $B_u = \Phi_u/\pi_u$ restores the exact filter regardless of which π was used to simulate (KLI Thm. 5, Lemma 3). Because Φ is a property of the target process alone, it is unchanged by which of the four kernels below is used; only π (or, for Hard, the auxiliary intensity κ) and the resulting importance weights change.

Notation. For a cross-deme transmission with source deme d write $I = I_d(y^-)$, $\ell = \ell_d(y^-)$, and let $r_b \geq 0$, $b \in \mathrm{col}_d(y^-)$, be the guide's *relative hazard* of recoloring branch b: the hazard of b's reverse-time coalescent transition relative to the hazard at stationarity, so $r_b \approx 1$ when the guide is uninformative and $\sum_b r_b = \ell$ exactly at stationarity. All three kernels mix a base law π_0 with the uniform floor

$$\pi_\varepsilon(j) = (1-\varepsilon)\pi_0(j) + \frac{\varepsilon}{\ell+1}, \qquad 0 \leq \varepsilon \leq 1, \quad j = 0, 1, \ldots, \ell,$$

already used for the naïve kernel above ($j = 0$ is the identity outcome, $j = 1, \ldots, \ell$ each a tracked branch), but the three kernels use three different base laws π_0. The code's `proposal_floor` parameter is this ε; as of v29, $\varepsilon = 0$ is legal and disables the floor, reducing π_ε exactly to π_0 (see the v29 changelog note).

Soft. Soft preserves the naïve aggregate split $\pi_0(0) = (I-\ell)/I$, $\sum_{j\geq 1} \pi_0(j) = \ell/I$, distributing the tracked mass over branches by relative hazard,

$$\pi_0(b) = \frac{\ell}{I} \cdot \frac{r_b}{\sum_{b'} r_{b'}} \qquad (b \in \mathrm{col}_d(y^-)),$$

falling back to the uniform share $1/\ell$ when $\sum_{b'} r_{b'} = 0$ (or is non-finite). Flooring the full $(\ell{+}1)$-vector gives

$$\pi_\varepsilon(b) = (1-\varepsilon)\frac{\ell}{I} \cdot \frac{r_b}{\sum_{b'} r_{b'}} + \frac{\varepsilon}{\ell+1},$$

whose tracked-group sum $\sum_{b\geq 1} \pi_\varepsilon(b)$ equals exactly the naïve $(1-\varepsilon)\ell/I + \varepsilon\ell/(\ell{+}1)$: the guide changes *which* tracked branch is proposed, never *how likely* the tracked group is as a whole.

Guided. Guided instead jointly normalizes the identity weight *together with* every branch's relative hazard, before flooring:

$$\pi_0(j) = \frac{w_j}{\sum_k w_k}, \qquad (w_0, w_1, \ldots, w_\ell) = (I-\ell,\, r_1, \ldots, r_\ell),$$

with the same uniform fallback $1/(\ell{+}1)$ when $\sum_k w_k = 0$, then $\pi_\varepsilon(j) = (1-\varepsilon)\pi_0(j) + \varepsilon/(\ell{+}1)$. Because the identity weight $I - \ell$ competes directly against every r_b in one normalization, a single dominant guide hazard can push $\pi_\varepsilon(0)$ far below the naïve ratio $(I-\ell)/I$ — the guide reallocates mass *between* identity and tracked outcomes here, not only *within* the tracked group as in Soft. This is what makes Soft and Guided mathematically distinct kernels rather than two implementations of the same law.

Hard. Hard uses the raw (unnormalized) relative hazards as *auxiliary intensity multipliers*, so the auxiliary process need not carry the same total rate as the population process:

$$\kappa_\varepsilon(0) = (1-\varepsilon)\frac{I-\ell}{I} + \frac{\varepsilon}{\ell+1}, \qquad \kappa_\varepsilon(b) = (1-\varepsilon)\frac{r_b}{I} + \frac{\varepsilon}{\ell+1} \quad (b \in \mathrm{col}_d(y^-)).$$

$\kappa_\varepsilon(0)$ is identical to Soft's/naïve's identity term, but $\sum_{b\geq 1} \kappa_\varepsilon(b)$ need *not* equal ℓ/I (equality holds only when the guide is at stationarity, $\sum_b r_b = \ell$). Writing α for the population-process rate of the transmission (e.g. $\beta_{HC} S_H I_C / N_C$), the auxiliary rates are $\beta_j = \alpha\, \kappa_\varepsilon(j)$, and the excess or shortfall relative to α is returned to the decay λ:

$$\Delta\lambda = \alpha - \sum_{j=0}^{\ell} \beta_j = \alpha\Big(1 - \kappa_\varepsilon(0) - \textstyle\sum_{b\geq 1} \kappa_\varepsilon(b)\Big),$$

which may be *negative* — Hard can propose color changes faster than the population process actually produces them, exactly the behavior its name describes. The realized importance correction for a drawn outcome j is

$$\log \frac{\alpha \Phi_j}{\beta_j} = \log \frac{\Phi_j}{\kappa_\varepsilon(j)},$$

so the α factor cancels and only the base-law difference from Soft/naïve enters the weight.

Aggregate-plus-conditional equivalence. Soft and Hard can each be realized as a two-stage draw — an aggregate "some tracked branch changes" event at rate $\alpha \sum_{b \geq 1} \pi_\varepsilon(b)$ (Soft) or $\alpha \sum_{b \geq 1} \kappa_\varepsilon(b)$ (Hard), followed by a conditional draw of the specific branch b with probability $\pi_\varepsilon(b) / \sum_{b' \geq 1} \pi_\varepsilon(b')$ (resp. $\kappa_\varepsilon(b) / \sum_{b' \geq 1} \kappa_\varepsilon(b')$) — rather than a single draw from the full per-outcome vector. The two stages recombine to the full per-outcome law exactly:

$$\underbrace{\Big(\textstyle\sum_{b' \geq 1} \pi_\varepsilon(b')\Big)}_{\text{aggregate}} \cdot \underbrace{\frac{\pi_\varepsilon(b)}{\sum_{b' \geq 1} \pi_\varepsilon(b')}}_{\text{conditional}} = \pi_\varepsilon(b),$$

so an implementation that applies the two log-corrections in sequence (once for the aggregate draw, once for the conditional branch) obtains exactly $-\log \pi_\varepsilon(b)$ — the same correction as a single draw from the full $(\ell+1)$-outcome law. The identical argument holds for κ_ε. §13 identifies where this two-stage structure appears in the code, and how it is checked.

Root and singular fork proposals. The root- and fork-coloring proposals shared by all three guided-family kernels mix the guide's forward present-probabilities with the same ε-floor, but *only* over the outcomes whose population rate is nonzero: at a fork, if (say) $S_C = 0$ then the same-deme outcome TCC is structurally infeasible and keeps probability exactly 0 regardless of ε, while every outcome with positive population rate keeps probability at least ε/n, for n the number of such feasible outcomes — so a vanishing guide weight can never assign zero proposal mass to a target-compatible fork or root outcome. (This is the defect the floor repairs: with an unfloored guide weight, $S_C > 0$ together with a guide belief of exactly 0 on TCC previously starved the particle filter of a valid coloring at that node.) The same construction applies to the root proposal, mixing only over demes with free (untracked) capacity.

11 Log-Likelihood

Following KLI Algorithm B1 (sequential importance sampling, no resampling), each particle carries a running weight $V^{(p)}$ and the estimator is

$$\hat{L} = \frac{1}{P} \sum_{p=1}^{P} V_T^{(p)}, \qquad \log V_T^{(p)} = -\log q\Big(X_0^{(p)}, Y_0^{(p)}\Big) + \sum_{\tau \in J_{\text{reg}}^{(p)}} \log B_{u_\tau}^{(p)}$$
$$+ \sum_{t_i \in \text{ev}(Z)} \log\big(A_i^{(p)}\, B_i^{(p)}\big) - \int_0^T \lambda\big(t, x^{(p)}(t), y^{(p)}(t)\big)\, dt,$$

so $\hat{L}$ is unbiased for L (KLI Lemma 3) and $\log \hat{L}$ estimates $\log L$ with a downward Jensen bias. Here:

- $J_{\mathrm{reg}}^{(p)}$ is every realized *regular* jump with $B_{u_\tau} \neq 1$, not only color-changing jumps: under the collapsed within-deme proposal (§9) a background TCC carries the full C-V boost $B_{\mathrm{TCC}} = 1 - \ell_C(\ell_C - 1)/[I_C(I_C - 1)] \neq 1$;

- $B_u = \Phi_u/\pi_u$, so $B_u = 0$ for an incompatible proposal and $B_u = 1$ *iff* $\pi_u = \Phi_u$ (compatibility alone does not give $B_u = 1$);

- at a data event t_i the increment is $\log(A_i B_i)$ *including the rate* A_i: for a sample/death leaf $A_i = \chi_d I_d(x')$ and $B_i = 1$, so the contribution is $\log[\chi_d I_d(x')]$, not zero.

The filter-equation identity $\beta B = \alpha\Phi$ does *not* remove the regular boosts from the particle weight, since a particle simulates from π and reweights by Φ/π (KLI Thm. 5, Lemma 3).

Initial coloring. Theorem 5 also requires a root-coloring proposal $q(x, y)$: with Y_0 latent, each particle draws $Y_0^{(p)} \sim q(X_0, \cdot)$ and carries the initial weight

$$V_0^{(p)} = \frac{1}{q\Big(X_0, Y_0^{(p)}\Big)}, \qquad w(0, x, y) = p_0(x)\, \mathbb{1}_{q(x,y)>0}.$$

The p_0 in $w(0, \cdot)$ is realized by sampling $X_0 \sim p_0$, not carried in the weight (had X_0 been drawn from a separate proposal g, $V_0 = p_0(X_0)/[g(X_0)\, q(X_0, Y_0)]$); thus $-\log q(X_0^{(p)}, Y_0^{(p)})$ enters $\log V_T^{(p)}$. A normalized, support-safe choice for k root lineages is the labeled (without-replacement) hypergeometric $q_x(y) = (I_C)_{\ell_C(y)}(I_H)_{\ell_H(y)}/(I_C{+}I_H)_k$, $k = \ell_C(y){+}\ell_H(y)$, with $(n)_j$ the falling factorial (no $\binom{k}{\ell_C}$ multiplicity, since that appears only when summing over distinct labeled colorings). It is defined on states with $I_C{+}I_H \geq k$ (otherwise $(I_C{+}I_H)_k = 0$; the numerator already vanishes unless $I_C \geq \ell_C(y), I_H \geq \ell_H(y)$); states unable to support the k root lineages carry zero weight. For the MERS tree the typed tips do not fix the internal/root colors, so this term is generally present; only if the root coloring is imposed externally is $q \equiv 1$ and $V_0 = 1$.

Mixed forks. At a typed leaf $\mathcal{U}_e$ is a singleton, so $A_i = \chi_d I_d(x')$. At an internal fork the feasible marks depend on perspective: for a fixed *mixed* post-event coloring the backward filter sums the predecessor components T_{HC} and T_{CH} (the parent color is latent); in the *forward* SMC each particle knows its pre-event parent color, so a camel parent permits $\{T_{CC}, T_{HC}\}$ and a human parent $\{T_{HH}, T_{CH}\}$. The total singular rate is $A_i = \sum_{u\in\mathcal{U}_e} \sum_{y^+} \int \beta^{\mathrm{ev}}_{e,u}\, dx^+$. The singular kernel proposes a feasible mark by its rate share and a post-event coloring through π_u (one outcome for a same-deme mark, two orientations for a cross-deme mark); A_i restores the total singular rate and $B_u = \Phi_u/\pi_u$ corrects the conditional coloring proposal.

12 Summary

1. **No decoupling.** THC and TCH span both demes; cross-deme branch points $\kappa^{bb'}_{CHC}, \kappa^{bb'}_{HCH}$ are possible.

2. **Recipe (§4):** (A) slots and ancestral demes; (B) saturations; (C) $\phi_u = \prod_d \binom{I_d - \ell_d}{r_d - s_d}/\binom{I_d}{r_d}$ with I_d, ℓ_d *post-event*; (D) operator $\sigma^b_{d_{\mathrm{pre}} d_{\mathrm{post}}}$ or $\kappa^{bb'}$.

3. **THC:** $r = (1, 1)$; the C-slot remains in C and contributes to the reduced identity term, while the H-slot is ancestrally in C despite landing in H (σ^b_{CH}); branch point $\kappa^{bb'}_{CHC}$. TCH is the mirror.

4. **Cross-color regular coefficient** uses the post-event count: $\frac{I_C-\ell_C(y)}{I_C I_H}$ (Eq. 10), $\frac{I_H-\ell_H(y)}{I_H I_C}$ (Eq. 12); no spurious -1.

5. **Sample/death:** $r = (0,0) \Rightarrow \phi = 1$; singular weight $\alpha_{S_d} = \chi_d I_d(x')$, no $1/I_d$ factor; only terminal leaves ($\chi_d^{a\emptyset}$).

6. **Decay** $\lambda = \chi_C I_C + \chi_H I_H + \gamma_d I_d \mathbb{1}_{I_d \le \ell_d}$. The reduced regular exit rate has births full, deaths above-threshold, and no sampling; the per-mark driver kernels are KLI's $\beta_u^{\text{reg}} = \alpha_u^{\text{reg}} \pi_u$ (Theorem 5).

7. **SMC weights** carry regular-jump contributions $\sum_{\tau \in J_{\text{reg}}} \log(\Phi_{u_\tau}/\pi_{u_\tau})$, singular rate–boost contributions $\sum_{t_i \in \text{ev}(Z)} \log(A_i B_i)$, and decay $-\int_0^T \lambda(t, X_t, Y_t)\, dt$.

13 From Math to Code: Implementation Map

This section is the only place in this document that names Julia files, functions, or tests; it exists to connect §10 to the actual implementation, deliberately placed after the derivation and the summary rather than alongside the math.

Shared code. `src/examples/mers_funs.jl` holds everything shared by all three guided-family kernels: `knowledge!` and `singular_part!` (the root- and fork-coloring proposals of the last paragraph of §10, and the shared `filter_pomp` constructor with its `proposal_floor` parameter, i.e. ε), together with the epsilon-floor building blocks used by the three kernels below: `floored_shares` and `floored_shares_feasible` implement $\pi_\varepsilon(j) = (1-\varepsilon)\pi_0(j) + \varepsilon/n$ over a feasible outcome set of size n; `cross_proposal` implements the naïve/Soft identity-vs-tracked-group split $(\pi_0(0), \sum_{j\ge1} \pi_0(j))$; `floored_branch_law` implements Soft's per-branch $\pi_\varepsilon(b)$; `hard_branch_law` implements Hard's $\kappa_\varepsilon(b)$.

Relation to this repository's SEIR implementation. The three guided-family kernels were designed by generalizing this repository's existing single-deme-swap SEIR implementation (`seir_soft.jl`, `seir_guided.jl`, `seir_hard.jl`, and their shared `seir_funs.jl`) to MERS's two-host, two-direction cross-deme structure, and by adding the ε-floor, which the SEIR implementation does not have at all (every SEIR kernel is support-*unsafe* in the same way the naïve MERS kernel was before this floor was added). Concretely:

- **Soft** (`mers_soft.jl`) mirrors `seir_soft.jl`'s architecture: an aggregate identity/tracked-group split (there, via `seir_funs.jl`'s generic `transmission!`, parameterized by `onI=ellI`) followed by a conditional branch draw via `PhyloPOMP.choose_branch(rh, n, cols, i, j)` (the low-level, precomputed-hazard overload in `guide.jl`, populated once per interval by `relhaz_alloc`/ `relhaz!`). MERS's version keeps the same two-stage shape but replaces the split and the conditional draw with the floored `cross_proposal`/`floored_branch_law` above, and generalizes from one swap direction to two (camel→human and human→camel).

- **Guided** (`mers_guided.jl`) mirrors `seir_guided.jl`'s architecture: a single combined transmission rate per direction, with the identity-vs-branch choice made in one joint draw via `PhyloPOMP.choose_branch(t, guide, node, n, cols, i, j)` (the higher-level overload in `guide.jl` that recomputes the hazard on demand). Because that function has no floor, and `guide.jl` is shared code that `seir_guided.jl`/`seir_hard.jl` depend on unfloored, `mers_guided.jl` does not edit it; instead it defines a local, floored analogue, `floored_choose_branch`, built from the shared `floored_shares` helper.

- **Hard** (`mers_hard.jl`) mirrors `seir_hard.jl`'s architecture: the same `seir_funs.jl` generic `transmission!`, but parameterized by `onI=sum_relhaz(...)` (the raw hazard sum, not the tracked-lineage count), and the same `choose_branch(rh, n, cols, i, j)` for the conditional branch draw. MERS's version replaces the unnormalized rate split with the floored `cross_proposal`/`hard_branch_law` above, and keeps the compensating-decay term (`decay` accumulating `rate - alpha[identity] - alpha[swap-group]`) that `seir_hard.jl` already uses for the same purpose.

One structural difference from the SEIR baseline: `seir_guided.jl` does not `include("seir_funs.jl")` (it duplicates `singular_part!` instead), whereas all three MERS kernels `include("mers_funs.jl")`, since the epsilon-floored root/fork proposal is shared identically across all three here.

Tests. `test/mers_soft.jl`, `test/mers_guided.jl`, and `test/mers_hard.jl` each check: construction, the zero-initial-infected $-\infty$ likelihood, simulation, and a finite-likelihood particle filter on a small hand-built fixture (the real 270-tip Dudas et al. genealogy used elsewhere in this document is too large for a modest particle count to reliably match every observed tip deme, so it is used only for the zero-infected and constructor checks). Beyond that:

- `test/mers_soft.jl` additionally checks $\ell = 0 \Rightarrow \pi_\varepsilon(0) = 1$; strict positivity of every outcome at the $I = \ell$ boundary; that $\sum_{b \geq 1} \pi_\varepsilon(b)$ from `floored_branch_law` agrees with `cross_proposal`'s tracked-group mass (the aggregate-plus-conditional equivalence, numerically); and that Soft's and Guided's log-likelihood estimates on the same genealogy and a deliberately nonuniform guide differ by more than Monte Carlo noise (the Soft-vs-Guided distinction argued in §10).

- `test/mers_hard.jl` additionally checks the same boundary/positivity properties for `hard_branch_law`, the aggregate-plus-conditional equivalence for κ_ε, and the rate/weight identity $\beta_j = \alpha\, \kappa_\varepsilon(j)$ together with $\log(\alpha \Phi_j / \beta_j) = \log(\Phi_j / \kappa_\varepsilon(j))$ from §10.

- `test/mers_guided.jl` checks the joint-normalization boundary/positivity properties directly on `floored_shares` (the building block of `floored_choose_branch`), and that a dominant relative hazard pushes the joint identity share below the fixed naïve ratio $(I - \ell)/I$, as §10 argues.

`test/mers_naive.jl` cannot follow this pattern: unlike the three guided-family kernels, `NaiveMERS.filter_pomp` (v28, resynced to upstream) takes no genealogy argument at all and always filters the full `mers_tree`, so there is no way to substitute a small fixture. Its tests are accordingly narrower: construction, the zero-initial-infected $-\infty$ check, the `Ic0/I_c0` and `Ih0/I_h0` alias-conflict `ArgumentError`s, and that `simulate/pfilter` run on the full tree without erroring — *not* that the resulting log-likelihood is finite, since the naïve kernel has no guide and matching every one of the tree's ~ 274 observed tip demes by chance is astronomically unlikely regardless of N_p (see §10's motivation for the other three kernels). Under current defaults ($\chi_H = 0$) the full-tree filter in fact always returns $-\infty$, at the first Human sample node, before guide quality or population dynamics are even relevant; see `mers_kernel_diagnostic_findings.md`.

All four test files pass under `RUN_HEAVY_TESTS=no julia --project=. test/runtests.jl`.

Benchmark results. With RUN_HEAVY_TESTS enabled, each of the three guided-family test files also runs a heavier particle-filter benchmark ($N_p = 1000$, 10 replicates) on the same small test genealogy and parameters used by the tests above, reporting the mean log-likelihood estimate $\log \hat{L}$ and its Monte Carlo standard error (via `logmeanexp`). `test/mers_naive.jl` has no equivalent benchmark as of v28: since `NaiveMERS.filter_pomp` cannot be pointed at the small fixture (see above), the only genealogy available to it is the full tree, on which it deterministically returns $-\infty$ under current defaults — `logmeanexp` over ten all-$-\infty$ replicates would itself collapse to `NaN`, which is why the naive heavy block was removed rather than left to silently regress. One such run of the remaining three:

Kernel	tests passed	wall time	$\log \hat{L} \pm \text{SE}$
Soft	21/21	17.1s	-107.9 ± 2.32
Guided	16/16	12.0s	-111.64 ± 0.684
Hard	17/17	16.8s	-111.9 ± 1.54

(Naive's earlier small-fixture figure, $\log \hat{L} = -114.63 \pm 0.358$, was measured before v28's API change removed the ability to filter a custom genealogy with the naïve kernel at all; it is no longer reproducible and is omitted rather than left looking current.) The three estimates cluster around the same value, as expected: the exact target likelihood $L(\theta)$ does not depend on the proposal kernel π (KLI Theorem 5, Lemma 3, and §10 above), only the Monte Carlo variance of $\hat{L}$ does. Guided has the tightest spread here (± 0.684) and Soft the widest (± 2.32); this is consistent with (but, from a single run, not strong evidence for) Guided's joint identity/branch reweighting concentrating proposal mass more effectively on data-compatible outcomes for this fixture. This table reports a single small-fixture verification run at a fixed seed and is reproducible via `julia --project=. test/runtests.jl` (default settings); it demonstrates internal consistency across these three kernels, not a scientific finding about the real MERS epidemic, and it should not be read as evidence that any one kernel is more efficient in general (only N_p=1000, 10 replicates, one fixture).

References

[1] King, A. A., Lin, Q., and Ionides, E. L. *Exact phylodynamic likelihood via structured Markov genealogy processes.* arXiv:2405.17032 (posted 2024, revised 2026).

[2] Dudas, G., Carvalho, L. M., Rambaut, A., and Bedford, T. (2018). MERS-CoV spillover at the camel-human interface. *eLife* 7:e31257.